

# Structured Memory Without Reusable Slots

How RNNs Represent Hierarchical Sequences

**36.8°**

**Hierarchical  
Organization**

**93%**

**Boundary  
Effect**

**0%**

**Slot  
Reusability**

**0.20**

**Chunk  
Modulation**

## **Conclusion:**

RNNs develop structure-specific representations  
consistent with Memory Buffers (El-Gaby et al.)